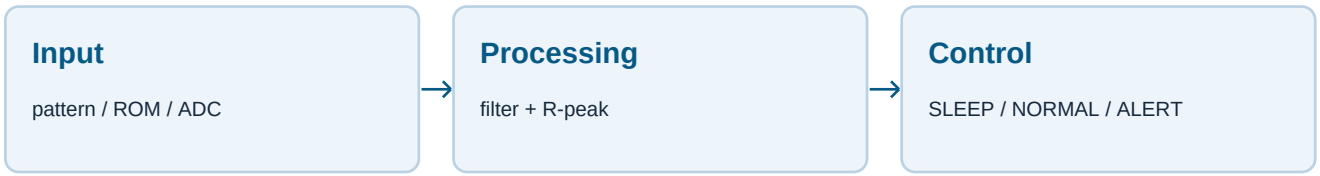
Adaptive ECG processing and clock gating

Portable digital demonstration for the Z15 / Zynq-7000 XC7Z015. This English technical report accompanies the public source package.

Design objective

The design stops processing-clock activity during intervals without useful input events, while retaining an always-on wake path. Baseline, adaptive-FSM-only and fully gated configurations separate the cost and behavior of the controller from the clock gate.



Always-on input/control observes activity; BUFGCE stops the processing clock during sleep.

Hardware and configuration

Item	Included configuration
Target and release clock	xc7z015clg485-2; 50 MHz; 20 ns constraint
Tool and HDL	Vivado 2025.2; Verilog/SystemVerilog; XADC IP for ADC variant
Core outputs	state, heart_rate, r_peak; physical release top exposes LED
ADC status	Q8 gain 5120 is a candidate setting, not calibrated analog gain

Replay modes

Mode	Input and matching release image
7	Internal synthetic activity sequence; mode7_release.bit (identical to gated_release.bit)
8	MIT-BIH record 119, MLII, 30-second replay; mode8_release.bit
9	MIT-BIH record 100, MLII, 30-second replay; mode9_release.bit

Complete AD8232 acquisition, calibrated analog gain and physical whole-board power measurement remain incomplete. This package demonstrates digital logic and archived evidence; it is not a clinical device.

Build and reproduce

Run commands from the public repository root. All scripts and data are local; the workflow does not create a remote repository or push files.

1 Check the package

```
python src/scripts/privacy_scan.py
python src/scripts/check_release.py
python src/scripts/bit_compare.py
```

The release manifest covers supplied file hashes. Mode 7 and gated release bitstreams must be byte-identical. Each included bitstream is approximately 3.51 MB, below the requested 50 MB inclusion threshold.

2 Run the digital regression

```
python src/scripts/run_tests.py
```

Requires Python 3, numpy, Icarus Verilog and vvp. There are 17 declared regression results. The included historical logs are evidence from earlier runs. Rechecking those logs is distinct from executing new simulations.

3 Rebuild release variants

```
vivado -mode batch -source src/scripts/build_all.tcl

vivado -mode batch -source src/scripts/build_vivado.tcl
-tclargs gated 5120
```

The second example is one command, wrapped here for readability. Substitute mode8 or mode9 to select the corresponding ROM input. Release checks require passing setup/hold timing, zero DRC errors and no ILA/VIO debug cores. Generated Vivado project/cache directories stay outside committed source.

4 Select a mode and configure the board

In Vivado Hardware Manager connect the matching Z15 target and select its mode7_release.bit, mode8_release.bit or mode9_release.bit from build/bitstreams/. Release mode selection is fixed by the chosen image. The existing programming helper performs volatile JTAG configuration only:

```
vivado -mode batch -source src/scripts/program_board.tcl
-tclargs mode7
```

The separate ecg_activity.bit debug image uses ecg_activity.ltx and VIO mode controls. It is not interchangeable with the release interface. Its rebuild script and archived validation are included under src/scripts/build_demo.tcl and build/demo_reports/.

Verification and evidence

Keep simulation, offline models and physical captures as separate evidence classes. Every percentage needs its own denominator and interval.

Regression coverage

The archived manifest contains 17 PASS results. Coverage includes ADC conditioning, XADC digital reading, integrated ADC paths, power variants, legacy adaptive behavior, modes 0..6, modes 8/9 and exact ROM-to-model peak comparisons.

Reproduction inputs: sim/ testbenches and src/scripts/run_tests.py. Saved results: build/reports/simulation/results.json. The ROM tests compare two complete loops and require a fixed pipeline offset rather than matching only an early subset of peaks.

Fresh native XSim comparison

Setting	Baseline	Gated
ENABLE_ADAPTIVE / USE_CLK_GATE	0 / 0	1 / 1
Input	Internal mode 7	Same internal mode 7
Steady-cycle clock edges	2,280,000	1,850,604
Input mismatches	0 across the paired run	0 across the paired run
Stable-sleep gated edges	Clock continues	0

The 22.8-second steady cycle yields 100 * (1 - 1850604 / 2280000) = 18.8331579%, rounded to 18.8%. The comparison testbench uses a 100 kHz functional clock with 1 kHz sampling to keep execution practical. It includes a zero-delay behavioral BUFGCE model, not routed device timing. Physical release builds use 50 MHz.

```
& ./sim/waveform_comparison/run_xsim.ps1
```

Run from a Vivado 2025.2 configured PowerShell environment. Native screenshots and recordings in report/waveforms/ display the same 18.000000 to 23.943141254463 second interval. Sleep entry is 18.703060 seconds and wake is 22.997060 seconds. Video clips show the saved native waveform display, not a live physical-board capture.

Archived physical evidence

The XADC summary records 4096 samples with code range 4..3033. A small representative mode 7 activity-counter capture is included under board/counters/. Windows crossing a state transition contain mixed activity; the state column is sampled at the end. Compute interval reduction from summed counters, not the mean of window percentages.

The approximately 18.28% historical board figure, 18.66% offline-model figure and 18.8% RTL figure have different scopes. No new physical-board measurement was performed during this assembly. See board/readme.md and docs/ad8232_status.md.

Resources, power and scope

Source numerical values are archived implementation reports. A modeled endpoint difference is not a measured board-power saving.

Archived release implementation values

Variant	LUT	FF	BRAM	DSP	Dyn. W	Total W
baseline	311	334	0.0	0.0	0.002	0.104
fsm_only	387	363	0.0	0.0	0.002	0.105
gated	389	364	0.0	0.0	0.003	0.105
adc	653	526	0.0	2.0	0.006	0.108
mode8	334	327	24.0	0.0	0.004	0.108
mode9	335	326	24.0	0.0	0.018	0.122

Baseline/FSM-only/gated comparisons use the same mode 7 chain. ROM and ADC variants add different hardware. The displayed common-default vectorless power values do not themselves establish a savings claim. Recalculate these values with src/scripts/summarize_reports.py from build/reports/.

Power claim boundary

The approximately 1.36 mW result is a controlled dynamic-power endpoint estimate from routed conditional vectorless CE=0/1 cases. There is no SAIF-driven full-workload measurement and no current-sensor reading. Confidence is low. Fixed-temperature/fixed-I/O release cases and their source reports are included under build/demo_reports/release_power/ and power_comparison.json. Debug/demo endpoints and static-power differences must be interpreted separately.

Data provenance and license

MIT-BIH Arrhythmia Database v1.0.0, Moody G and Mark R, PhysioNet (2005), DOI 10.13026/C2F305. Mode 8 uses record 119 from 1551.350 s; mode 9 uses record 100 from 31.724 s; each window lasts 30 s. Existing data were resampled to 1 kHz, DC removed, centered at 1000 counts, scaled at 1500 counts/mV and clipped to 0..4095.

Code uses MIT. The derived replay data retain the separate Open Data Commons Attribution License v1.0 notice in data/readme.md and docs/data_sources_and_license.md. Source link: https://physionet.org/content/mitdb/1.0.0/

AI collaboration and public release

ai_records/ represents both supplied Codex and Claude material with redacted public selections and source notes. These are historical collaboration records, not proof that every assistant claim was correct. Full private account exports are excluded. skill/ provides the current main workflow plus build, verification, board and power guidance.

The public checklist is docs/included_checklist.md. Filenames are lowercase; non-AI text is English. The release manifest records file hashes. The original engineering archives remain unchanged. The original Chinese technical PDFs were replaced here with this newly compiled English technical report.